

PAPER_939: Centaurus A Jet Power Curves

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** blazar_jet_power_curves_extended.py (CentaurusAJetPowerCurves) **Calculator:** CentaurusAJetPowerCurvesCalc (CP4 #523) **CVW:** v2.0.0 compliant

Abstract

We compute numerical jet power curves for Centaurus A (NGC 5128), the nearest radio galaxy at $d \approx 3.8$ Mpc. Using the Blandford-Znajek mechanism with UQFF phonon linewidth modulation, we derive power enhancement factors at three linewidths $\Gamma = 0.05, 0.10, 0.30$ THz. The SMBH parameters are $M_{\text{BH}} = 5.5 \times 10^7 M_{\odot}$, $a = 0.70$, $B = 3000$ T, and $A_{\text{jet}} = 0.95$, yielding enhancements of $2.6 \times / 2.1 \times / 1.4 \times$ respectively.

1. System Parameters

Parameter	Value	Source
M_{BH}	$5.5 \times 10^7 M_{\odot}$	EHT/VLBI
Spin a	0.70	Jet morphology fits
B	3000 T	Faraday rotation
A_{jet}	0.95	Phonon coupling
Distance	3.8 Mpc	Tully-Fisher

2. Blandford-Znajek Power

$$P_{\text{BZ}} = \frac{B^2}{8\pi} \left(\frac{r_H}{c} \right)^2 a^2 c$$

where $r_H = \frac{r_S}{2}(1 + \sqrt{1 - a^2})$ and $r_S = 2GM/c^2$.

3. UQFF Jet Modulation

$$M_{\text{jet}}(\Gamma) = 1 + A_{\text{jet}} \exp\left(-\frac{(\Gamma - \Gamma_0)^2}{2\sigma_{\Gamma}^2}\right)$$

$$P_{\text{jet}}(\Gamma) = P_{\text{BZ}} \cdot (1 + M_{\text{jet}}(\Gamma))$$

Γ (THz)	Enhancement
0.05	$2.6 \times$
0.10	$2.1 \times$
0.30	$1.4 \times$

4. Source Data

- **File:** blazar_jet_power_curves_extended.py
 - **Session:** 213
 - **CP4 Class:** CentaurusAJetPowerCurvesCalc (#523)
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References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Israel, F.P. (1998) – Centaurus A, A&AR, 8, 237
3. Blandford, R.D. & Znajek, R.L. (1977) – MNRAS, 179, 433